

Research Article

Multimodal Control of Vortex-Induced Vibration of a Long-Span Suspension Bridge Using MR Dampers

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Due to their high flexibility and low damping, long-span suspension bridges are susceptible to multimodal vortex-induced vibration (MVIV) under low and normal wind speeds. However, it remains a challenge for the currently used control strategies to achieve optimal additional damping ratios for different modes of vibration as wind speed varies. To this end, this study presents a multimodal control strategy for mitigating MVIV of long-span suspension bridges using magnetorheological (MR) dampers. A vortex-induced force (VIF) model is first established based on the VIFs identified from the wind and structural responses of a bridge during MVIV measured on site. The MVIV of the bridge is then simulated by applying the VIF model to the finite element model of the bridge, and the optimized setup of the control system, consisting of MR dampers and supporting brackets, is sought in terms of a passive control strategy. The multimodal control strategy, which is a novel semiactive control strategy, is finally proposed based on the self-excited characteristics of MVIV observed on site and a linear quadratic regulator. To demonstrate the effectiveness and robustness of the proposed control strategy, a real long-span suspension bridge once suffering MVIV is chosen as a case study. The results demonstrate that the proposed control strategy can robustly mitigate the MVIV of the bridge in the first fourteen modes of vibration in vertical direction, and the effectiveness of the proposed strategy is superior to passive or other semiactive control strategies.

Keywords: long-span suspension bridge; magnetorheological damper; multimodal vortex-induced vibration; semiactive control

1. Introduction

A long-span suspension bridge often has dynamic characteristics of low natural frequencies and damping ratios in vertical direction. Within a low and normal wind speed range, the bridge may experience vortex-induced vibration (VIV) in one of the first few vertical modes of vibration as incoming wind speed varies [1, 2], leading to the so-called multimodal vortex-induced vibration (MVIV). Over the years, MVIV has been observed on several long-span bridges. For instance, the Xihoumen suspension bridge experienced the vertical MVIV from the 2nd to 9th modes of vibration within a wind speed range of 4–18 m/s [1], while the Humen suspension bridge encountered the MVIV in

three different vertical modes of vibration within a wind speed range of 4–12 m/s [2]. Although MVIV is not as catastrophic as the flutter, excessive vortex-induced response (VIR) will have a negative impact on the driving safety and comfort [3, 4] as well as the fatigue performance of structural components [5]. Therefore, it is crucial to find an effective and implementable countermeasure to mitigate MVIV of a long-span suspension bridge.

Methods for mitigating VIV of long-span suspension bridges typically include structural measures, aerodynamic countermeasures, and damping devices [6]. Among those, the structural measures by modifying either the stiffness and/or mass of the bridge to change its natural frequencies often encounter challenges such as high cost and

ineffectiveness. Aerodynamic countermeasures are often considered by finding an optimal aerodynamic configuration of the bridge deck or adding some aerodynamic devices to the bridge deck during the design stage of the bridge through wind tunnel tests [7, 8]. However, due to the complex nature of MVIV, aerodynamic countermeasure may not be universally effective and may be difficult to be implemented if the bridge is in operation [9]. The installation of damping devices, such as nonlinear energy sink [10] and tuned mass dampers (TMD), into the bridge deck is a common practice for VIV control [11–13]. Due to the broadband vibration energy absorption ability, the nonlinear energy sink can be adopted to mitigate MVIV [10]. TMDs absorb vibration energy by adjusting their natural frequency to the structural frequency of VIV [14]. However, due to the relatively low natural frequencies of the bridge deck, the dynamic displacements of TMDs often exceed the allowable space inside the deck, making it challenging to use TMDs to mitigate VIV. Furthermore, the primary aim of TMD parameter optimization design is typically focused on only one specific mode of vibration, which results in the suboptimal or no control of other modes of vibration. Huang et al. [15] demonstrated a novel strategy for determining TMD parameters to minimize the modal coupling effects and ensure effective MVIV control. To control MVIV, multiple sets of TMDs of different natural frequencies are required [16], resulting in the excessive increase of dead loads on the bridge deck. Therefore, the installation of TMDs for MVIV control remains a challenging issue.

There has been a growing trend recently on the use of damped outriggers for VIV control of long-span suspension bridges. Damped outriggers offer a significant advantage by providing supplementary damping for the bridge, considering the substantial deformation of the bridge deck during VIV. The effectiveness of damped outriggers has been substantiated through rigorous analytical and numerical analyses on the optimal locations and parameters of damped outriggers and the damping coefficients of the viscous dampers [17, 18]. However, it is noted that the aforementioned studies concerned only the additional damping ratios from the damped outriggers to specified modes of vibration, and they did not investigate the actual impact on the mitigation of VIR or quantify the effectiveness in terms of VIR reduction. Furthermore, the selected optimal damping coefficient for one specified mode of vibration may not guarantee enough additional damping ratios for other modes of vibration. If additional damping ratios are required for several modes of vibration, the large sizes of viscous dampers in the damped outriggers sometime are needed. Therefore, the feasibility and effectiveness of using damped

outriggers for MVIV control also require further investigation.

A MR damper is a semiactive control (SAC) device which cannot inject mechanical energy into the controlled bridge system but has adjustable properties to optimally reduce the response of the bridge system. There are several commonly used SAC strategies for the control of MR dampers [19–21], but they are often used for vibration control of buildings and bridges under wind or earthquake excitation, in which the dynamic properties of the structures will not be changed by external forces. Vortex-induced force (VIF) of a long-span suspension bridge is a self-excited vibration resulting from the interaction of bridge motion with incoming wind. When a long-span suspension bridge is subjected to VIV, VIF will alter the dynamic properties of the bridge and make the bridge deviate from its original dynamic state. The traditional SAC algorithms do not consider such self-excited forces.

To address the limitations of the existing MVIV control methods, the present study aims to propose a novel SAC strategy for mitigating MVIV of a long-span suspension bridge using magnetorheological (MR) dampers with adjustable damping forces. The remainder of this paper is organized as follows. Section 2 provides all the key elements required in the development of the MVIV control strategy, which include equations of VIF model, MR damper model, and semiactive MVIV control strategy. Section 3 presents a case study to demonstrate the feasibility, effectiveness, and robustness of the proposed control strategy, in which a real long-span suspension bridge and its MVIV events, modelling of the bridge and VIF, MVIV analysis of the bridge, thresholds for MVIV of the bridge, arrangement of control system, optimal damping coefficient of passive control, SAC versus passive control, and comparison with traditional SAC strategies are introduced in sequence. The main conclusions from this study are summarized in Section 4.

2. Key Elements in Proposed Control Strategy

The key elements required in the development of the MVIV control strategy include VIF model, MR damper model, and semiactive MVIV control strategy.

2.1. VIF Model. Jiang et al. [22] recently proposed a new framework for nonlinear VIF identification and modelling of a long-span bridge using the wind and structural response data recorded on site during VIV events. The proposed nonlinear VIF model for VIV of a bridge deck in the vertical direction is used in this study and can be expressed as

$$F_{VIV} = \frac{1}{2}\rho U^2 (2D) \left[Y_1 \frac{\dot{x}}{U} + Y_2 \frac{x}{D} - Y_1 \varepsilon_1 \left(\frac{\dot{x}}{U} \right)^3 - Y_1 \varepsilon_2 \left(\frac{\dot{x}}{U} \right)^2 \frac{x}{D} - Y_1 \varepsilon_3 \frac{\dot{x}}{U} \left(\frac{x}{D} \right)^2 - Y_1 \varepsilon_4 \left(\frac{x}{D} \right)^3 \right], \quad (1)$$

where F_{VIV} represents the VIF acting on the unit length of the bridge deck; x is the vertical displacement response of the bridge deck at one point; the over-dot denotes differentiation with respect to time; ρ is the air density; U is the mean wind speed; D is the characteristic length of the bridge deck; $Y_1, Y_2, \varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4$ are the six parameters in the VIF model, determined by the field measured wind and structural response data for a given bridge.

As shown in equation (1), VIF is the function of the displacement and velocity of a bridge deck, which will change the dynamic characteristics of the bridge. As a result, to mitigate VIV of the bridge, it is crucial to consider the self-excited characteristics of VIF. Furthermore, VIF is also the function of mean wind speed, which will excite VIV of the bridge in different modes of vibration. Therefore, it is also important to consider the control device which can mitigate VIV of the bridge for different modes of vibration.

2.2. MR Damper Model. A MR damper consists of mainly MR fluid, rod, magnetic choke, accumulator, piston, and wires to electromagnet, as shown in Figure 1 [23–25]. The damping force $F_{MR\text{ DAMPER}}$ provided by the MR damper can be calculated by the modified Bouc–Wen hysteresis model [23, 24]. Figure 2 shows a schematic diagram of the modified Bouc–Wen hysteresis model for the damping force of a MR damper. The governing equation of the damping force can be expressed as

$$F_{MR\text{ DAMPER}} = \alpha z_D + c_0(\dot{x}_D - \dot{y}_D) + k_0(x_D - y_D) + c_1\dot{y}_D + k_1(x_D - x_0), \quad (2)$$

where the evolutionary variable z_D and internal displacement y_D of the damper are governed by

$$\dot{z}_D = -\gamma|\dot{x}_D - \dot{y}_D|z_D|z_D|^{n-1} - \beta(\dot{x}_D - \dot{y}_D)|z_D|^n + A(\dot{x}_D - \dot{y}_D), \quad (3)$$

$$\dot{y}_D = \frac{(\alpha z_D + k_0(x_D - y_D) + c_0\dot{x}_D)}{(c_0 + c_1)}, \quad (4)$$

where x_D is the displacement response of the rod of the MR damper; $F_{MR\text{ DAMPER}}$ represents the damping force generated by the MR damper and it is an element in the damping force vector $F_{MR\text{ DAMPER}}$ in equation (1); k_1 represents the

accumulator stiffness; k_0 represents the control stiffness at a high velocity; c_0 represents the viscous damping at a high velocity; c_1 represents the viscous damping at a low velocity; x_0 represents the initial displacement of the MR damper; α, β, γ, n , and A are the five parameters of the MR damper. The above-mentioned parameters α, c_0 , and c_1 are the functions of voltage u and can be expressed by

$$\begin{aligned} \alpha &= \alpha_a + \alpha_b u, \\ c_0 &= c_{0a} + c_{0b} u, \\ c_1 &= c_{1a} + c_{1b} u, \end{aligned} \quad (5)$$

where $\alpha_a, \alpha_b, c_{0a}, c_{0b}, c_{1a}, c_{1b}$ are the constant coefficients.

2.3. Semiactive MVIV Control Strategy. Based on the traditional linear quadratic regulator (LQR) control algorithm, this study proposed a SAC strategy considering self-excited forces, abbreviated as SAC-SE, whose target is to mitigate VIV through the control of the MVIV of the bridge in the state space.

In terms of the conventional finite element (FE) method, the equations of motion for a n -degree of freedom bridge system with r MR dampers subjected to VIV can be expressed as

$$M\ddot{X} + C\dot{X} + KX = B_F F_{VIV} + B_s F_{MR\text{ DAMPER}}, \quad (6)$$

where M, C , and K are the mass, damping, and stiffness matrices of the bridge including the supporting brackets for the MR dampers, respectively; X is the displacement response vector of the bridge; F_{VIV} is the vector of VIFs acting on the bridge deck; $F_{MR\text{ DAMPER}}$ is the vector of damper forces provided by the MR dampers; B_s is the $n \times n$ position matrix of damper forces; B_F is the $n \times n$ position matrix of VIF; and the over-dot denotes the differentiation with respect to time.

To best describe the dynamic state of a bridge subjected to VIV, the VIF expressed by equation (1) is divided into a linear VIF and a nonlinear VIF. The linear VIF comprises only the linear aeroelastic damping term and the linear aeroelastic stiffness term, while the nonlinear VIF is the rest part of the VIF expressed by equation (1). Accordingly, the linear VIF vector can be expressed by equation (7) and moved to the left side of equations (6), and (6) can be rewritten as equation (8).

$$F_{VIV_linear} = H_1 \dot{X} + H_2 X, \quad (7)$$

$$M\ddot{X} + (C - H_1)\dot{X} + (K - H_2)X = B_F F_{VIV_nonlinear} + B_s F_{MR\text{ DAMPER}}, \quad (8)$$

where F_{VIV_linear} is the linear VIF vector; $H_1 = Y_1 B_F$ is the matrix of linear negative aeroelastic damping force; $H_2 = Y_2 B_F$ is the matrix of linear negative aeroelastic stiffness force; and $F_{VIV_nonlinear}$ is the nonlinear VIF vector.

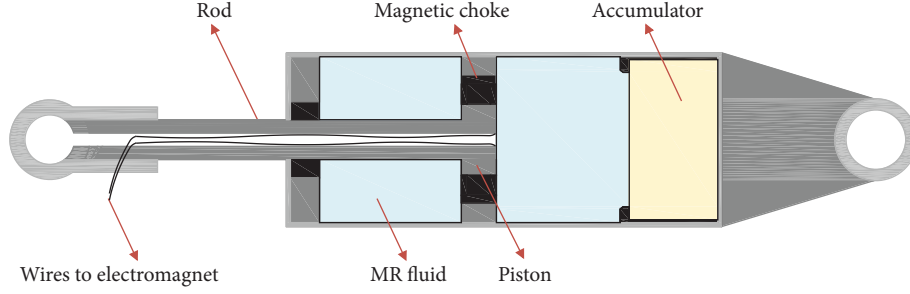


FIGURE 1: Basic configuration of a MR damper.

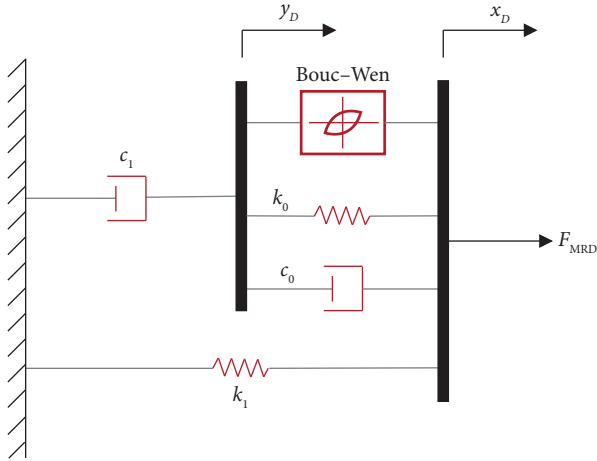


FIGURE 2: Schematic diagram of modified Bouc-Wen hysteresis model for MD damper.

The state-space form of equation (8) can be expressed as

$$\dot{Z} = AZ + BF_{MR\text{ DAMPER}} + DF_{VIV_nonlinear}, \quad (9)$$

where $Z = \begin{bmatrix} X \\ \dot{X} \end{bmatrix}$, $A = \begin{bmatrix} 0_n & I_n \\ -M^{-1}(K - H_2) & -M^{-1}(C - H_1) \end{bmatrix}$, $B = \begin{bmatrix} 0_n \\ M^{-1}B_s \end{bmatrix}$, $D = \begin{bmatrix} 0_n \\ M^{-1}B_F \end{bmatrix}$. In which I_n is the $n \times n$ identity matrix; 0_n is the $n \times n$ zero matrix; and the character n denotes the number of degrees of freedom of the bridge system.

The key advantage of LQR control is its ability to simultaneously consider optimality and stability. Building upon equation (9), one can describe the performance objective function as

$$J = \int_0^\infty [Z^T Q Z + F_{MR\text{ DAMPER}}^T R F_{MR\text{ DAMPER}}] dt, \quad (10)$$

where $Q = \alpha_{SAC_SE} \begin{bmatrix} K - H_2 & 0_n \\ 0_n & M \end{bmatrix}$ and $R = \beta_{SAC_SE} I_n$ are the weighting matrices in the performance objective function of the SAC-SE control. α_{SAC_SE} and β_{SAC_SE} are the coefficients of weight matrices Q and R , respectively. The SAC-SE

control refers to the SAC strategy proposed in this study. For the traditional SAC-LQR control, $Q = \alpha_{SAC_LQR} \begin{bmatrix} K & 0_n \\ 0_n & M \end{bmatrix}$ and $R = \beta_{SAC_LQR} I_n$, where α_{SAC_LQR} and β_{SAC_LQR} are the coefficients.

The weight matrices will have an impact on the damper force of MR dampers and the response reduction of the controlled structure. In practical application, the values of α_{SAC_SE} and β_{SAC_SE} are generally determined by trial and error to balance the response reduction and the control energy required. The minimum value of the performance objective function is taken according to the minimum principle, and the state feedback gain matrix G can be expressed by equation (11).

$$G = R^{-1} B^T P, \quad (11)$$

where P is a $2n \times 2n$ matrix, which can be calculated by the algebraic Riccati equation [26]:

$$-PA - A^T P + PBR^{-1}B^T P - Q = 0. \quad (12)$$

According to the solution of algebraic Riccati equation and the minimum value of the performance objective function, the state feedback gain matrix G can be determined and the optimal control force of MR damper can be calculated by

$$F_{MR\text{ DAMPER}}^{\text{opt}} = -GZ. \quad (13)$$

It can be seen from equations (11) to (13) that A and Q play crucial roles in calculating the state feedback gain matrix G and the optimal control force of MR damper, whereas A and Q involve linear VIFs, as shown in equations (9) and (10), leading to MR damper interacted with VIFs. Since MR damper is a semiactive damper, the optimal control force $F_{MR\text{ DAMPER}}^{\text{opt}}$ obtained by LQR cannot be directly used as the output of the MR damper. It is necessary to change the applied voltage to make the output of the MR damper meet the requirements of the optimal control force as much as possible. Thus, selecting an appropriate control strategy based on optimal damping force is crucial. This study adopts the following equation as the SAC control of the MR damper.

$$F_{\text{MR DAMPER}} = \begin{cases} F_{\min}, & \text{others,} \\ F_{\text{MR DAMPER}}^{\text{opt}}, & F_{\text{MR DAMPER}}^{\text{opt}} \nu < 0 \text{ and } |F_{\min}| < |F_{\text{MR DAMPER}}^{\text{opt}}| < |F_{\max}|, \\ F_{\max}, & F_{\text{MR DAMPER}}^{\text{opt}} \nu < 0 \text{ and } |F_{\text{MR DAMPER}}^{\text{opt}}| > |F_{\max}|, \end{cases} \quad (14)$$

where $F_{\min}$ is the minimum control force of the MR damper according to current state; $F_{\max}$ is the maximum control force of the MR damper according to current state; ν is the relative velocity between the two ends of the MR damper.

When implementing the proposed SAC strategy in practice, it is necessary to know the incoming mean wind speed and the acceleration responses of the bridge deck from the anemometers and accelerometers installed on the bridge. From the measured acceleration responses, the VIV mode of vibration can be determined, and the responses at both ends of the MR damper can be obtained. Then, according to the incoming mean wind speed and the identified VIV mode of vibration, the corresponding coefficients Y_1 and Y_2 can be determined by the methodology described in Section 3.2.2, and the values of A and B in equation (9) can be computed. Then, utilizing A , B , and the corresponding Q and R , the state feedback gain matrix G can be derived. Consequently, based on the state feedback gain matrix G , the velocity and displacement of MR damper and corresponding coefficients Y_1 and Y_2 , the proposed SAC strategy is implemented for the designated VIV mode of vibration. If the incoming mean wind speed changes and another VIV mode of vibration is excited, H_1 and H_2 also change because the matrices of H_1 and H_2 are related to the incoming mean wind speed due to the vortex-induced linear force, so as the matrices of A , B , Q , R , and G . Therefore, the controller shall be changed semi-actively for another VIV mode of vibration. Clearly, leveraging the field measurement data from the structural health monitoring (SHM) system installed in the bridge can effectively enforce the proposed SAC strategy. Figure 3 illustrates the MVIV control framework for a long-span suspension bridge using MR dampers, which will be further elaborated in detail in the subsequent case study.

3. Case Study

3.1. A Real Suspension Bridge and VIV Events

3.1.1. Description of the Bridge. A real long-span suspension bridge is selected as the prototype bridge to demonstrate the feasibility, effectiveness, and robustness of the proposed MVIV control strategy. The bridge has a total length of 2713 m with a span arrangement of 578 + 1650 + 485 m, as shown in Figure 4(a).

The north side span is supported by suspenders, and the south side span of the bridge is supported by a series of piers. Since the bridge is located at a typhoon-prone area, a twin-box steel deck is used for wind resistant design. The deck has a cross section of 36 m in width and 3.51 m in height, and the gap between the two box girders is 6 m. The bridge deck carries two-way road traffic, each way having two lanes. More details on the cross section can be found in Figure 4(b). Compared with the single-box deck, the twin-box deck is

more intricate, rendering the bridge more susceptible to MVIV occurrences. Field measurement data from the SHM system installed on the bridge have revealed several MVIV events of the bridge [1]. Therefore, it is of great practical significance to mitigate MVIV of the prototype bridge.

3.1.2. Observed MVIV Events. On June 18, 2020, with the SHM system installed on this bridge, three VIV events were observed and recorded. Figure 5 presents the measured vertical acceleration response time history of the main span of the bridge at the 1/4 section as well as the corresponding three normalized power spectral density (PSD) functions. It is revealed from these figures that the first VIV event occurred with a dominant frequency of 0.325 Hz, the second VIV event with a dominant frequency of 0.275 Hz, and the third VIV event with a dominant frequency of 0.276 Hz. These observations confirm that this bridge indeed experienced MVIV with the successive occurrence of three VIV events. It is also interesting to observe that apart from the dominant frequency in the PSD of each event, the PSD also had other frequencies with considerable vibration energy. It is thus imperative to implement a suitable multimodal control strategy for this bridge to effectively mitigate MVIV of the bridge.

3.2. Modelling of Bridge and VIV

3.2.1. FE Model of Bridge. The updated three-dimensional (3D) FE model in [22] is directly adopted in this study. This FE model was established using the software ANSYS, as shown in Figure 6.

The twin-box deck was idealized as the two parallel longitudinal beams connected by cross beams at 3.6-m intervals. The longitudinal beams, cross beams, and towers were simulated using 3D elastic beam elements with tension, compression, torsion, and bending capacities. The main cables and hangers were modelled with cable elements. The main cables are anchored at both ends and fixed at the top of the two towers without slip. The main beams are constrained in the transverse, vertical, and longitudinal rotation directions at the south tower and the north end of the bridge. The main beams were constrained in the transverse and longitudinal rotation directions at the north tower. Furthermore, longitudinal dampers are set at the south tower and the north end of the bridge for longitudinal restraint. However, in consideration that this study primarily concerns the vibration control of vertical VIV and longitudinal damping devices have minimal impact on vertical vibration, longitudinal rods are used to substitute longitudinal damping devices in this study. Nevertheless, the damping matrix (C) of the bridge was constructed by Rayleigh damping assumption based on the measured damping ratio

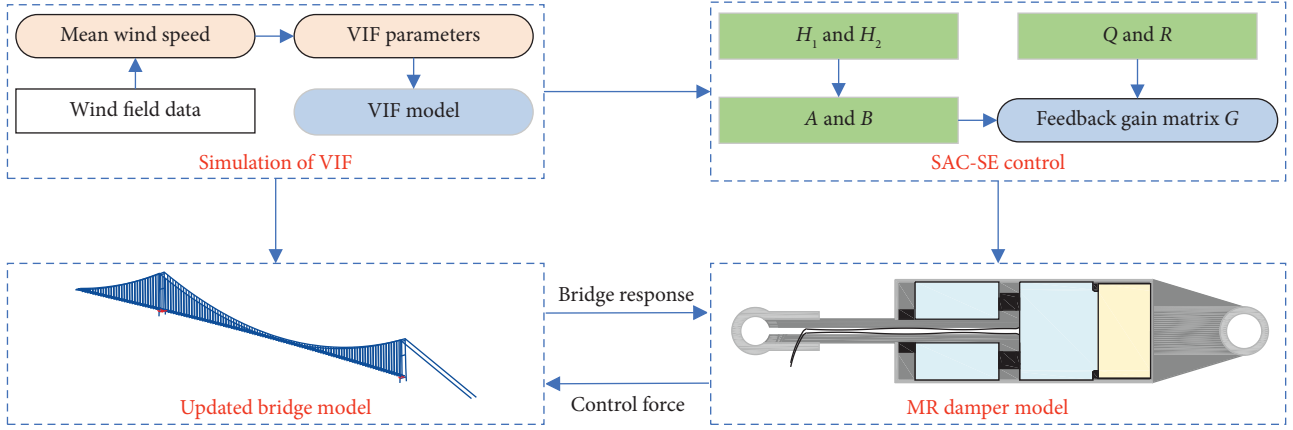


FIGURE 3: MVIV control framework for a long-span suspension bridge using MR dampers.

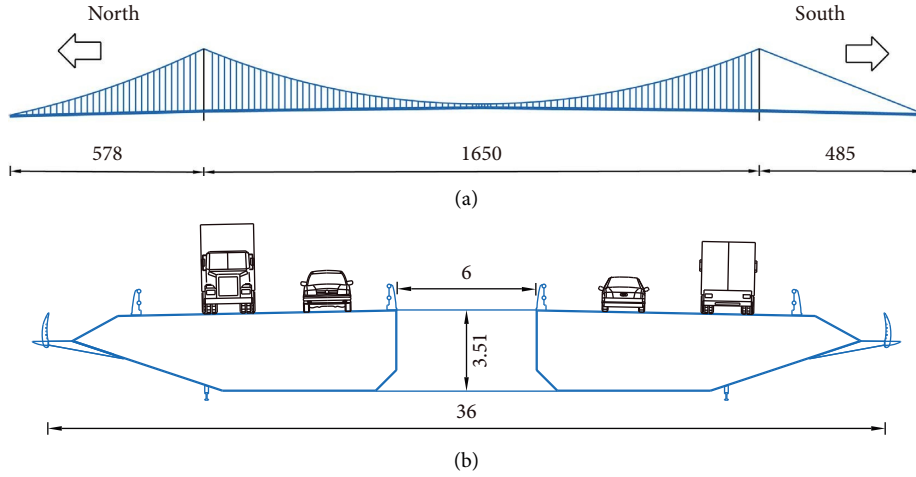


FIGURE 4: Prototype bridge: (a) elevation view of the bridge (m); (b) cross section of bridge deck (m).

of 0.57% in the 1st vertical mode and damping ratio of 0.5% in the 14th vertical mode. Therefore, if the longitudinal damping devices could attribute minimal damping to the vertical modes of vibration, the measured damping ratios in the vertical modes of vibration have already included it. The FE model had a total of 2242 nodes and 3136 elements. The mass matrix (M), stiffness matrix (K), and damping matrix (C) of the bridge can be therefore extracted from the established FE model for this case study. For further details on the FE model of the prototype bridge, please refer to [22].

3.2.2. VIF Model. As mentioned in Section 2.1, the VIF model described in reference [22] is adopted in this study. The VIF parameters identified for the prototype bridge at the reduced wind speed U_r of 9.177 are presented in Table 1. In the six VIF parameters of the VIF model, ε_1 , ε_2 , ε_3 , and ε_4 are the parameters for the nonlinear VIF, Y_1 is the parameter for the linear aeroelastic damping force, and Y_2 is the parameter for the linear aeroelastic stiffness force. The reduced wind speed here is defined as $U_r = U/fD$, where f is the natural frequency of a given mode of vibration, U is the mean wind speed, and D is the depth of the deck. It is noted that as mean

wind speed increases, higher modes of VIV will occur but the reduced wind speed remains unchanged. Therefore, the VIF parameters listed in Table 1 remain the same for different modes of VIV. It is also noted that when the reduced wind speed U_r is 9.177, the vortex-induced displacement amplitude of the bridge deck reaches its maximum. Therefore, to better assess the feasibility and effectiveness of the MVIV control strategy, all the MVIVs are analyzed at the reduced wind speed U_r of 9.177 in the case study.

3.3. MVIV Analysis of Bridge. The MVIV analysis of the prototype bridge without additional dampers is carried out in this section. According to the reference [1], the MVIV of the bridge was observed to occur in the vertical modes of vibration at the frequencies of 0.183, 0.230, 0.276, 0.327, 0.378, 0.435, and 0.493 Hz, with the highest vertical mode being the 13th vertical mode. Since the occurrence of VIV in an even higher vertical mode requires an even higher mean wind speed but has a very low probability, the vertical modes of vibration of the prototype bridge with the frequencies below 0.54 Hz will be considered for further analysis in this case study. The modal frequencies and mode shapes of the

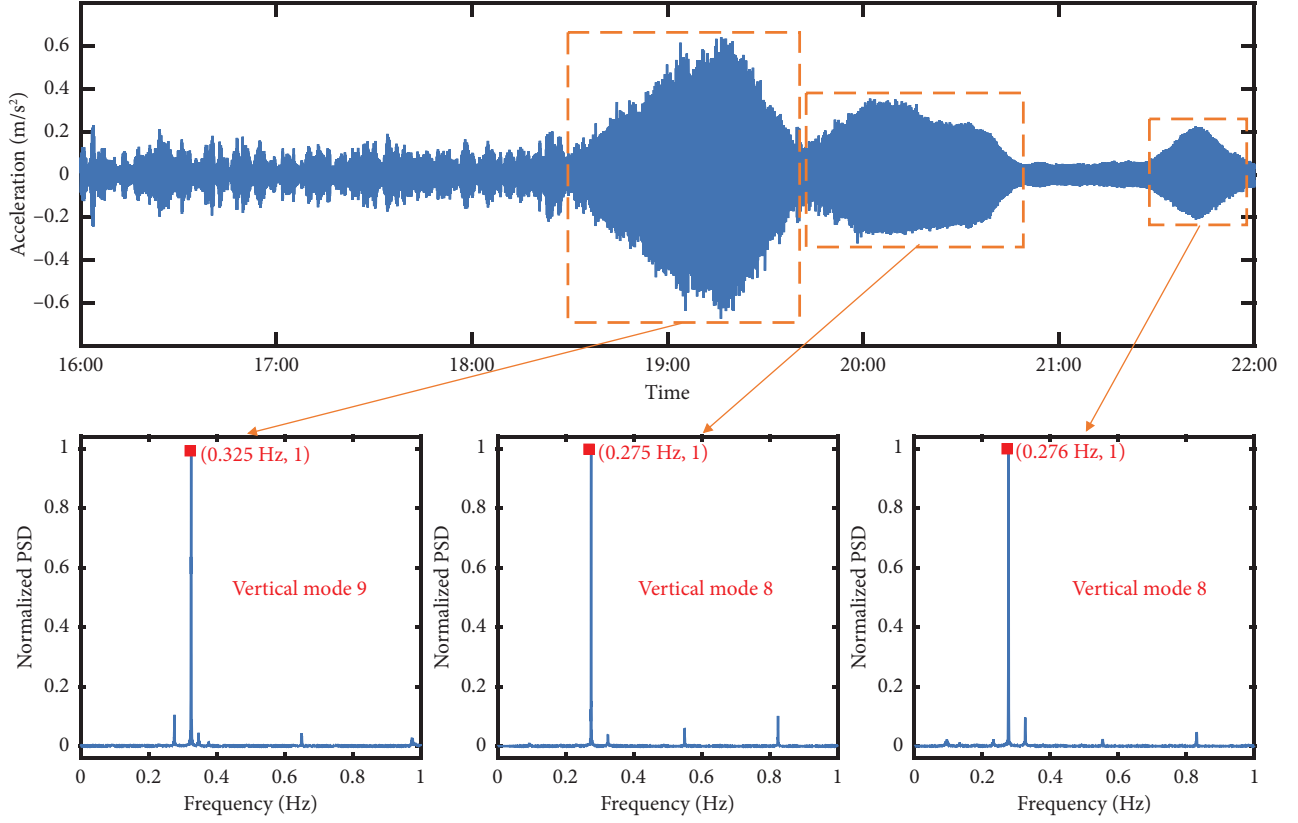


FIGURE 5: Measured vertical acceleration response time history of MVIV of the bridge and corresponding PSD functions.

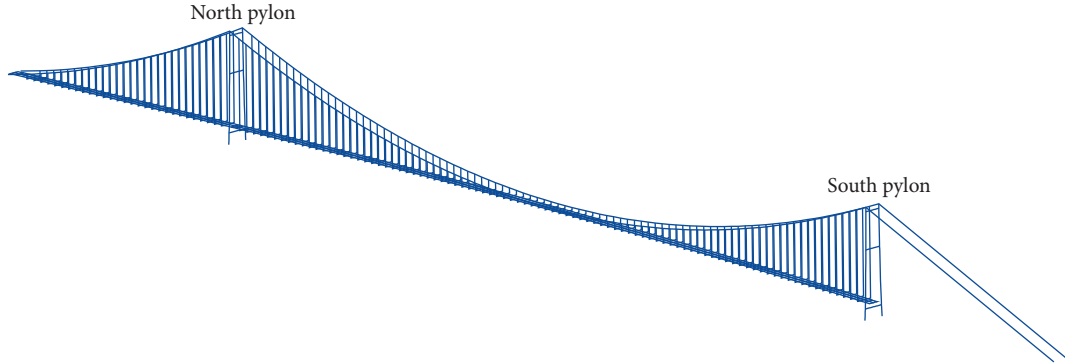


FIGURE 6: Three-dimensional FE model of the prototype bridge.

TABLE 1: VIF parameters in the VIF model for the prototype bridge [22].

Reduced wind speed	Y_1	Y_2	ϵ_1	ϵ_2	ϵ_3	ϵ_4
9.177	30.53	-37.89	-334.54	174.91	-709.99	369.01

first fourteen vertical modes of vibration of the prototype bridge are computed and presented in Figure 7. The frequencies in Figure 7 are the undamped frequencies calculated from the 3D FE model. Furthermore, the computed frequencies are also compared with the measured frequencies, as detailed in Table 2. The comparison revealed

a high level of consistency between the computed frequencies and the measured frequencies for the seven observed vertical modes of vibration in the field, thereby establishing a solid foundation for subsequent multimodal vibration control. According to the reference [17], the measured damping ratios of 1st, 3th, 5th, 6th, 8th, 9th, 12th and 13th are listed in Table 2. The vortex-induced maximum displacement amplitude of the bridge deck is also calculated for each of the first fourteen vertical modes of vibration at the reduced wind speed of 9.177 by using equation (6) without control. The maximum displacement amplitudes and the corresponding mean wind speeds for each of the first fourteen vertical modes of vibration are also listed in Table 2.

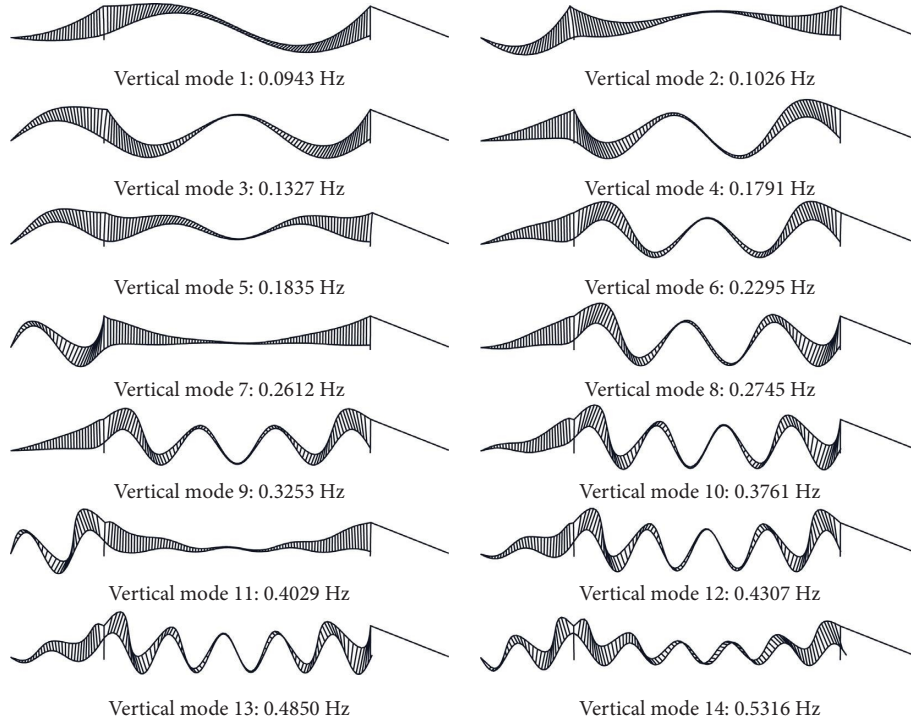


FIGURE 7: The first fourteen vertical bending modes of vibration of the prototype bridge without dampers.

TABLE 2: Natural frequencies, damping ratios, and maximum amplitudes of the bridge in vertical modes of vibration.

Vertical mode	Measured frequency (Hz)	Computed frequency (Hz)	Measured damping ratio (%)	Mean wind speed (m/s)	Maximum amplitude (m)
1st	—	0.094	0.57	3.019	0.252
2nd	—	0.103	—	3.308	0.250
3rd	—	0.133	0.52	4.272	0.254
4th	—	0.179	—	5.749	0.243
5th	0.183	0.183	0.50	5.878	0.273
6th	0.230	0.230	0.51	7.387	0.258
7th	—	0.261	—	8.383	0.238
8th	0.276	0.275	0.39	8.833	0.242
9th	0.327	0.325	0.42	10.439	0.234
10th	0.378	0.376	—	12.077	0.246
11th	—	0.403	—	12.944	0.249
12th	0.435	0.431	0.37	13.844	0.254
13th	0.493	0.485	0.32	15.578	0.270
14th	—	0.532	—	17.088	0.342

3.4. Threshold for VIV of a Long-Span Bridge. To online decide when the proposed MVIV control strategy should be applied, it is essential to know the threshold for MVIV of a long-span suspension bridge. However, there are different thresholds provided by different standards or codes, posing significant variations. The International Standard Organization (ISO) in ISO 2631-137 [27] uses the overall vibration total value (OVTV) and the motion sickness incidence (MSI) to evaluate the ride comfort. However, both OVTV and MSI values require the offline calculation of VIR time histories of the bridge deck.

Wind-resistant design specification for highway bridges (JTG/T 3360-01-2018) [28] is a crucial specification that outlines the guidelines for defining the MVIV displacement

amplitude threshold in China. The MVIV displacement amplitude threshold defined in this specification can be expressed as equation (15). It is noted that this threshold in this specification is limited to bridges with span lengths shorter than 200 m. However, there is currently no established threshold for VIV of long-span bridges. Therefore, this study adopts equation (15) as one evaluation criterion. The K-factor method [29] is additionally employed as another indicator in this study, with the minimum value between the two criteria serving as the final threshold.

$$h_{\text{JTG}} = \frac{0.04}{f_b}, \quad (15)$$

where h_{TG} is the MVIV displacement amplitude threshold; and f_b is the natural (vortex-shedding) frequency of the bridge deck.

The K-factor method has been extensively studied over the past several decades, and it remains a valuable tool for assessing the driving comfort of long-span suspension bridges. Based on the amplitude and frequency of structural vibration, the K-factor threshold value can be calculated by

$$\begin{cases} K = h_K f_b^2, & f_b < 5 \text{ Hz}, \\ K = h_K f_b, & 5 \text{ Hz} < f_b < 40 \text{ Hz}, \\ K = 200h_K, & 40 \text{ Hz} < f_b < 200 \text{ Hz}, \end{cases} \quad (16)$$

where h_K is the threshold value provided by the K-factor method; K is the K-factor, and the K value for assessing the sensitivity of the human to vibration is shown in Table 3.

In consideration that the total length of the prototype bridge is 2713 m, the estimated time for a vehicle to pass the bridge is approximately 2-3 min. Therefore, the K value of 10 in the K-factor method is adopted in this study. The threshold value for MVIV of the prototype bridge used in this study is therefore decided as the smaller value between the two threshold values h_{TG} and h_K . The values of h_{TG} and h_K as well as the threshold values used in this study for the first fourteen vertical modes of vibration of the bridge deck are calculated and listed in Table 4. It is noted from Table 4 that for the 1st to 6th vertical modes of vibration, the value of h_{TG} is smaller than the value of h_K . Conversely, for the higher vertical modes of vibration, the value of h_K is smaller than the value of h_{TG} . Furthermore, the maximum displacement amplitudes of the bridge deck calculated in the last section are also listed in Table 4. It is seen that only the threshold values for the first three vertical modes of vibration are larger than the corresponding maximum displacement amplitudes of the bridge deck. Therefore, it is necessary to adopt an effective countermeasure to mitigate the MVIV of the prototype bridge in other higher modes of vibration.

3.5. Arrangement and Initial Assessment of Control System

3.5.1. Control System. In this study, the layout of the control system is shown in Figure 8, in which a bracket is installed on the bridge tower horizontally to extend the MR damper support toward the mid-span. The MR damper is vertically positioned on the bracket with another end connected to the longitudinal beam of the bridge deck so that a relatively large vertical displacement can be produced at the MR damper position, resulting in a good control performance. It is worth noting that the bracket-viscous damper system was already used to mitigate VIV of the Yingwuzhou Yangtze River Bridge [30]. Furthermore, as indicated in reference [17], when the stiffness of the outrigger reaches a certain threshold, the additional damping ratio provided by the damper will no longer be affected by further increases in the stiffness of the outrigger. The same principle applies to the brackets used in this study. Once the bracket's stiffness reaches a sufficient level, it no longer influences the

TABLE 3: Human sensitivity to vibration.

K	Sensitivity to vibration
0.1	Perceptible vibrations (level 1)
1	Endurance for prolonged durations (level 2)
10	Limited tolerance for short durations (level 3)
100	Maximum vibration tolerance (level 4)

effectiveness of the control system. Therefore, a sufficiently high stiffness of brackets is adopted in this study to avoid the impact of stiffness of the bracket on the control effectiveness of the MR damper.

3.5.2. Arrangement of Control System. For the prototype bridge, there are three possible locations where the proposed control system can be installed, as shown in Figure 9. The control systems at the three possible locations are indexed as damper-bridge system 1, damper-bridge system 2, and damper-bridge system 3, respectively. The damper-bridge system 1 is installed on the left side of the north pylon. The damper-bridge system 2 is installed at the right side of the north pylon. The damper-bridge system 3 is installed at the left side of the south pylon. For the proposed control system, the bracket length plays an important role in controlling MVIV of a long-span suspension bridge. A short bracket length will hamper the effectiveness of MVIV control while a long bracket length will raise construction difficulty and cost. As a result, a 6-m bracket length is selected in this study according to the bracket-viscous damper system installed on a long-span suspension bridge in reference [30] to balance control effectiveness and cost. Moreover, the results from this study do demonstrate that a 6-m bracket can provide enough vertical displacement at the MR damper position and control effectiveness.

In order to better compare the control effects of three different damping systems, this section considers the passive viscous damper instead of the semiactive MR damper in the proposed control system and then adopts the dynamic complex analysis method [31, 32] to find the additional damping ratios ζ provided by the viscous dampers in different modes of vibration. The optimal damping coefficients and the corresponding maximal damping ratios provided by the three different damping systems for the first 14 modes are listed in Table 5. For 1st, 3rd, 4th, 6th, 8th, 9th, and 10th modes, the maximal damping ratios provided by the damper-bridge system 3 are greater than those by the damper-bridge system 1 or the damper-bridge system 2. However, the optimal damping coefficients of the damper-bridge system 3 are larger than those of the damper-bridge system 1 or the damper-bridge system 2 for almost all modes of vibration. Larger optimal damping coefficients means larger dampers and higher costs required. Nevertheless, in consideration of different scenarios of application, this study considers two combinations of damper-bridge systems, one is the more economical combination (damper-bridge systems 1&2) and the other is the more effective combination (damper-bridge systems 2&3).

TABLE 4: Threshold values for the first fourteen vertical modes of vibration of the bridge.

Vertical mode	h_{JTG} (m)	h_K (m)	Maximum amplitude (m)	Threshold (m)
1st	0.424	1.125	0.252	0.424
2nd	0.390	0.951	0.250	0.390
3rd	0.302	0.568	0.254	0.302
4th	0.223	0.312	0.243	0.223
5th	0.218	0.297	0.273	0.218
6th	0.174	0.190	0.258	0.174
7th	0.153	0.147	0.238	0.147
8th	0.146	0.133	0.242	0.133
9th	0.123	0.095	0.234	0.095
10th	0.106	0.071	0.246	0.071
11th	0.099	0.062	0.249	0.062
12th	0.093	0.054	0.254	0.054
13th	0.082	0.043	0.270	0.043
14th	0.075	0.035	0.342	0.035

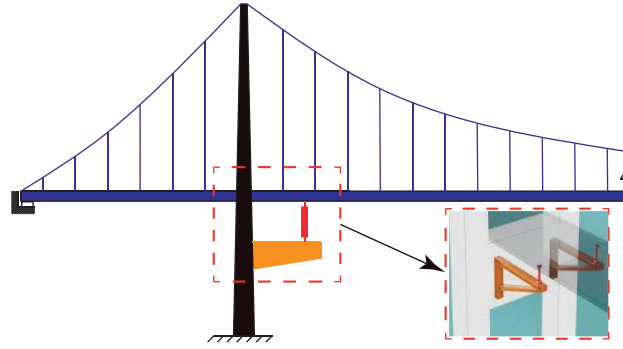


FIGURE 8: Layout of the control system with bracket and MR damper.

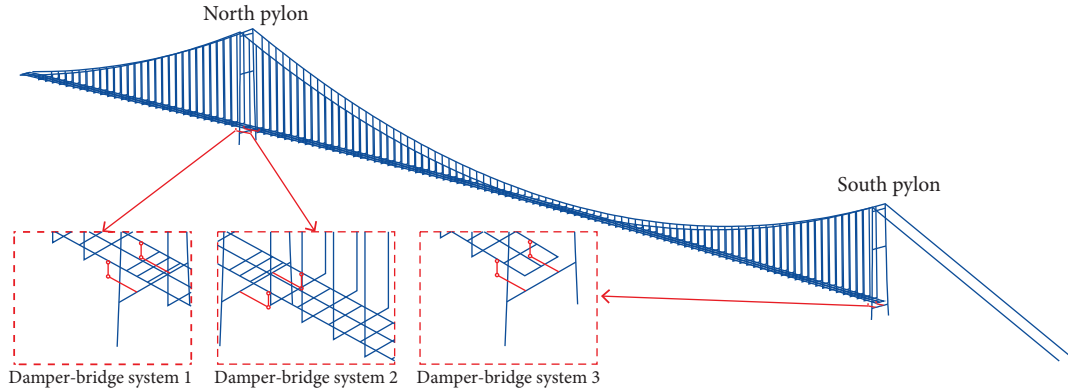


FIGURE 9: 3D FE model of the prototype bridge and three possible locations of the control system.

3.5.3. Initial Assessment of Control System. For the two damper-bridge systems, two viscous dampers of the same damping coefficients are used. The variations of the additional damping ratios provided by the damper-bridge systems 1&2 and the damper-bridge systems 2&3 in the first fourteen vertical modes of vibration are computed and plotted in Figures 10 and 11 against the damping coefficients of the viscous dampers. As shown in Figures 10 and 11, for the fourteen vertical modes of vibration, the additional damping ratios increase with the increasing damping coefficient of the dampers at the beginning, but once the

damping coefficient of the dampers exceeds a certain value, the additional damping ratio decreases as the damping coefficient of the dampers further increases. Therefore, there exists an optimal damping coefficient for the dampers to achieve the maximum additional damping ratio in a particular vertical mode of vibration. The maximum additional damping ratio and the corresponding optimal damping coefficient of the dampers, however, vary considerably for different modes of vibration.

Table 6 lists the optimal damping coefficients and the corresponding maximum additional damping ratios

TABLE 5: Multimodal damping of Xihoumen Bridge with damping systems.

Vertical mode	Optimal damping coefficient (N·s/m)			Maximal damping ratio (%)		
	Damper-bridge System 1	Damper-bridge System 2	Damper-bridge System 3	Damper-bridge System 1	Damper-bridge System 2	Damper-bridge System 3
1st	6.62×10^8	4.57×10^7	1.65×10^8	0.38	0.80	1.20
2nd	2.39×10^7	2.76×10^7	5.00×10^9	1.33	0.62	0.13
3th	2.00×10^8	9.74×10^7	1.19×10^8	1.08	1.09	1.17
4th	3.08×10^7	1.61×10^7	8.42×10^7	0.37	0.94	1.43
5th	2.13×10^7	4.79×10^7	5.00×10^9	0.36	0.13	0.12
6th	1.55×10^7	1.08×10^7	6.43×10^7	0.36	0.90	1.37
7th	1.03×10^7	1.67×10^7	6.70×10^8	3.04	1.31	0.05
8th	9.60×10^6	7.85×10^6	5.71×10^7	0.47	1.08	1.48
9th	8.45×10^6	6.35×10^6	4.81×10^7	0.55	1.18	1.49
10th	4.35×10^6	4.25×10^6	4.36×10^7	0.69	1.36	1.48
11th	4.60×10^6	5.30×10^6	1.15×10^8	4.01	2.07	0.11
12th	7.00×10^6	4.55×10^6	3.92×10^7	0.80	1.50	1.50
13th	3.05×10^6	2.85×10^6	3.72×10^7	1.09	1.82	1.48
14th	1.65×10^6	2.35×10^6	4.17×10^7	2.28	3.04	1.03

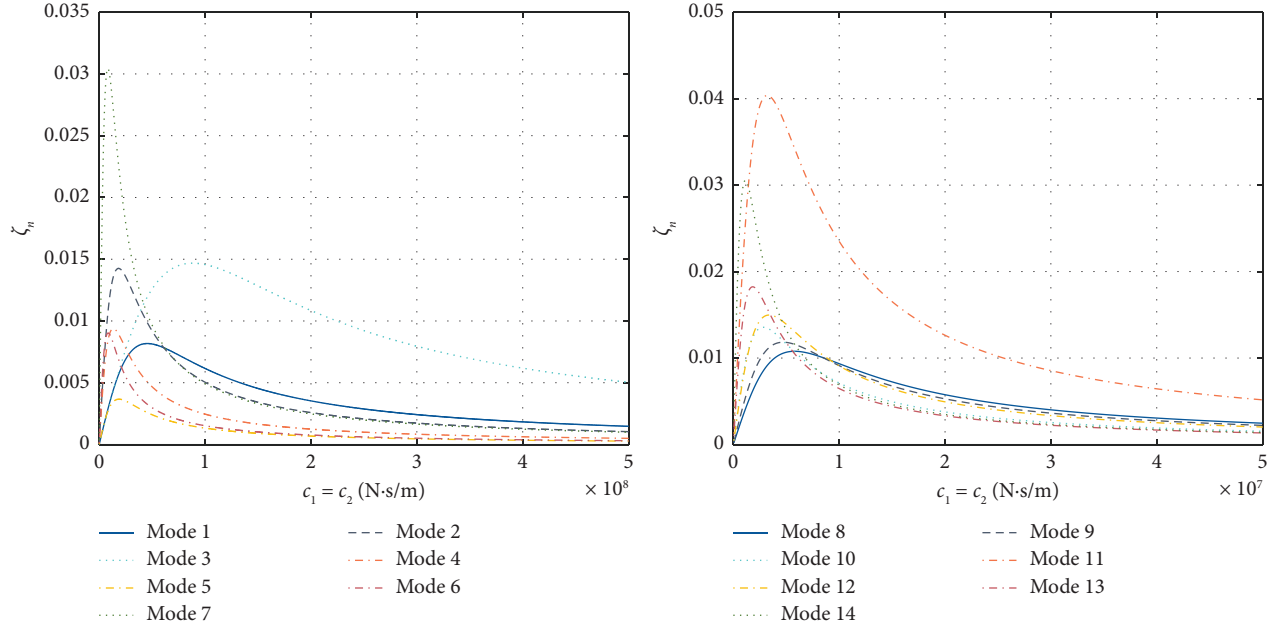


FIGURE 10: Damping curves of the prototype bridge with damper-bridge Systems 1 and 2.

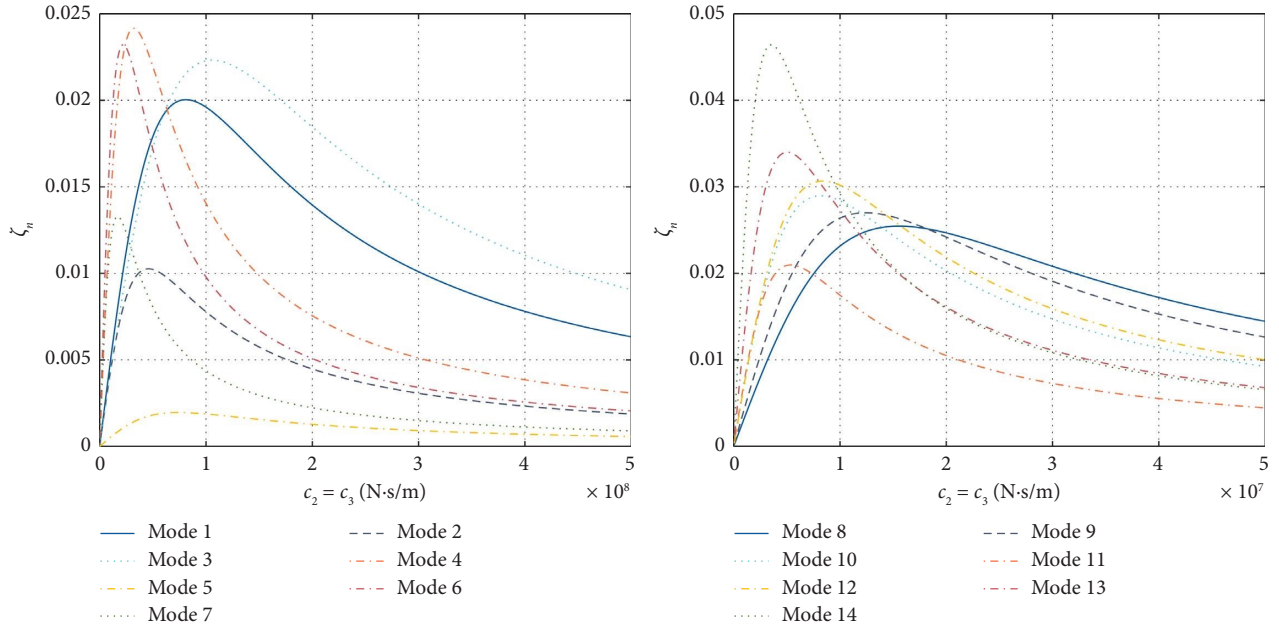


FIGURE 11: Damping curves of the prototype bridge with damper-bridge Systems 2 and 3.

provided by the damper-bridge systems 1&2 and the damper-bridge systems 2&3. It can be seen that the maximum additional damping ratios of the damper-bridge systems 2&3 are larger than those of the damper-bridge systems 1&2 for 1st, 3rd, 4th, 6th, 8th, 9th, 10th, 12th, 13th, and 14th vertical modes of vibration. For the 4th mode of vibration, the maximum additional damping ratio provided by the damper-bridge systems 2&3 is 2.57 times that of the damper-bridge systems 1&2. However, the optimal damping coefficient of the damper-bridge systems 2&3 is 2.2 times that of the damper-bridge systems 1&2. It is noted from Table 6

that if the viscous damper-bridge systems are installed, the maximum additional damping ratios in all the modes of vibration, except for 5th mode, are larger than 0.82%. This indicates that the proposed damper-bridge systems, regarding the positions of the dampers and the lengths of the brackets, are acceptable.

3.6. Suboptimal Damping Coefficient of Passive Control. By analyzing the maximum additional damping ratios in the first fourteen vertical modes of vibration listed in Table 6, it is evident that for each mode of vibration, the

TABLE 6: Maximum additional damping ratios provided by the viscous damper-bridge systems.

Vertical mode	Damper-bridge Systems 1 and 2			Damper-bridge Systems 2 and 3		
	Optimal damping coefficient of the dampers (N·s/m)	Maximum additional damping ratio (%)	Optimal damping coefficient of the dampers (N·s/m)	Maximum additional damping ratio (%)	Optimal damping coefficient of the dampers (N·s/m)	Maximum additional damping ratio (%)
1 st	4.54×10^7	0.82	8.11×10^7	2.00		
2 nd	1.83×10^7	1.43	4.59×10^7	1.03		
3 rd	8.83×10^7	1.47	1.06×10^8	2.23		
4 th	1.34×10^7	0.94	3.21×10^7	2.42		
5 th	1.87×10^7	0.37	7.44×10^7	0.20		
6 th	8.50×10^6	0.90	2.22×10^7	2.32		
7 th	8.15×10^6	3.05	1.69×10^7	1.33		
8 th	5.80×10^6	1.08	1.56×10^7	2.55		
9 th	4.70×10^6	1.18	1.24×10^7	2.70		
10 th	2.85×10^6	1.36	8.20×10^6	2.90		
11 th	3.20×10^6	4.04	5.35×10^6	2.10		
12 th	3.40×10^6	1.50	8.40×10^6	3.07		
13 th	1.85×10^6	1.82	5.05×10^6	3.40		
14 th	1.15×10^6	3.06	3.55×10^6	4.64		

optimal damping coefficient of the dampers leading to the maximum additional damping ratio is different. This feature makes it difficult to determine an optimal damping coefficient of viscous dampers in practice to obtain maximum additional damping ratio for the first fourteen vertical modes of vibration. Furthermore, the modes of vibration with a relatively low modal damping ratio require a relatively large additional damping ratio to ensure that all the fourteen vertical modes can obtain sufficiently high damping ratios. In view of this, this study proposes a simple way to determine a suboptimal damping coefficient of the dampers for all the modes of vibration: it is determined by maximizing the targeted expectation function R as expressed by equation (17).

$$R(c_i) = \sum_{j=1}^{14} \left(w_j \frac{\zeta_a^j(c_i) + \zeta_m^j}{\zeta_{\max}^j + \zeta_m^j} \right), \quad (17)$$

where w_j represents the weight coefficient of the j^{th} vertical mode of vibration; ζ_a^j represents the additional damping ratio in the j^{th} vertical mode of vibration provided by the damping coefficient c_i of the dampers; $\zeta_{\max}^j$ represents the maximum additional damping ratio that can be provided to the j^{th} vertical mode of vibration; ζ_m^j represents the modal damping ratio in the j^{th} vertical mode of vibration. For 2nd, 4th, 7th, 10th, 11th, and 14th vertical modes, due to lack of measured modal damping ratios, the modal damping ratios of these modes are assumed as 0.5%.

The weight coefficients in equation (17) are important for identifying the suboptimal damping coefficient for all the fourteen vertical modes of vibration. According to the field measurement data of the prototype bridge, VIV of the bridge was observed to occur in the 5th, 6th, 8th, 9th, 10th, 12th, and 13th vertical modes of vibration. Therefore, if the weight coefficient for one of these seven modes is 1 and the weight coefficients for all other modes are zero, the suboptimal damping coefficient of the damper-bridge systems 1&2 calculated by equation (17) will be the same as the optimal damping coefficient for this mode of vibration listed in Table 6, and the total damping ratios in all the fourteen modes of vibration can be calculated and listed in Figure 12. It can be observed from Figure 12 that when the damping coefficient of the dampers is the optimal damping coefficient for a specific mode, this specific mode can achieve the maximum total damping ratio but the other vertical modes of vibration cannot achieve their maximum total damping ratios.

In view of this contradiction and in considering the actual occurrence of VIV of the prototype bridge, this study assigns the weight coefficients of 2/21 to these seven field-observed vertical modes of vibration, while the weight coefficients for the other seven modes of vibration are set at 1/21. Then, by calculating the value of R with the damping coefficient ranging from 0 to 1×10^8 N·s/m, the relationship between the value of R and the damping coefficient can be sought by equation (17). Figure 13 shows that the suboptimal damping coefficients of the damper-bridge systems 1&2 and the damper-bridge systems 2&3 are 4.55×10^6 and

1.17×10^7 N·s/m, respectively, and the target expectation function R reaches the maximum value of 0.82 and 0.81, respectively.

3.7. SAC vs. Passive Control. Based on the results of optimal damping coefficients of the passive viscous dampers listed in Table 6, the four MR dampers of the maximum damper force of 200 kN are selected for the damper-bridge systems 1&2 with the two MR dampers installed at the location of the damper-bridge system 1 and the other two installed at the location of the damper-bridge system 2. For the damper-bridge systems 2&3, the eight MR dampers of the maximum damper force of 200 kN are selected with the four MR dampers installed at the locations of the damper-bridge system 2 or the damper-bridge system 3. The major parameters of the large-scale 200 kN MR dampers from reference [33] used in this case study are given in Table 7.

To assess the effectiveness and robustness of the proposed SAC strategy for MVIV of the bridge, this section compares the control performance between the passive control, two traditional SAC algorithms, and the proposed SAC in terms of the VIR reduction of the bridge in the specified fourteen modes of vibration. These two traditional SAC algorithms are bang-bang SAC algorithm and the semiactive LQR control algorithm (SAC-LQR). The VIR time histories of the bridge with the passive and SAC are calculated for a period of 400 s at an interval of 0.02 s.

Figures 14 and 15 display the VIR time history of the bridge deck at the maximum modal amplitude of each mode of vibration with different control methods. To compare the effectiveness of different control methods, the vortex-induced maximum displacement amplitudes of the bridge deck at the reduced wind speed of 9.177 in Table 2 are selected as the steady-state amplitudes of all the control methods. The results with the passive control are represented by the black line, with the bang-bang control by the blue line, and with the SAC-LQR control by the red line. The green lines represent the SAC-SE control, which is based on the self-excited characteristics of VIV in each mode of vibration. Furthermore, the variation of vibration amplitude of the bridge with different control methods and damper-bridge systems are listed in Table 8.

As shown in Figure 14, it is noted that the reductions of VIR with passive control in the 1st, 3rd, 4th, 5th, and 6th modes of vibration are very small. For the 5th mode of vibration, the passive control can only mitigate the VIR amplitude to 0.265 m, which exceeds the threshold value of 0.218 m. For the 6th mode of vibration, the passive control can only mitigate the VIR amplitude of the bridge deck to 0.183 m, which is still larger than the threshold value of 0.174 m. For the 9th and 10th modes of vibration, the passive control can mitigate the VIR amplitudes to 0.106 m and 0.111 m, respectively, but these values are still larger than the thresholds of 0.095 m and 0.071 m listed in Table 4. On the other hand, for the 2nd and 7th to 14th modes, the passive control can reduce the VIR amplitude significantly. Clearly, the passive control using the viscous

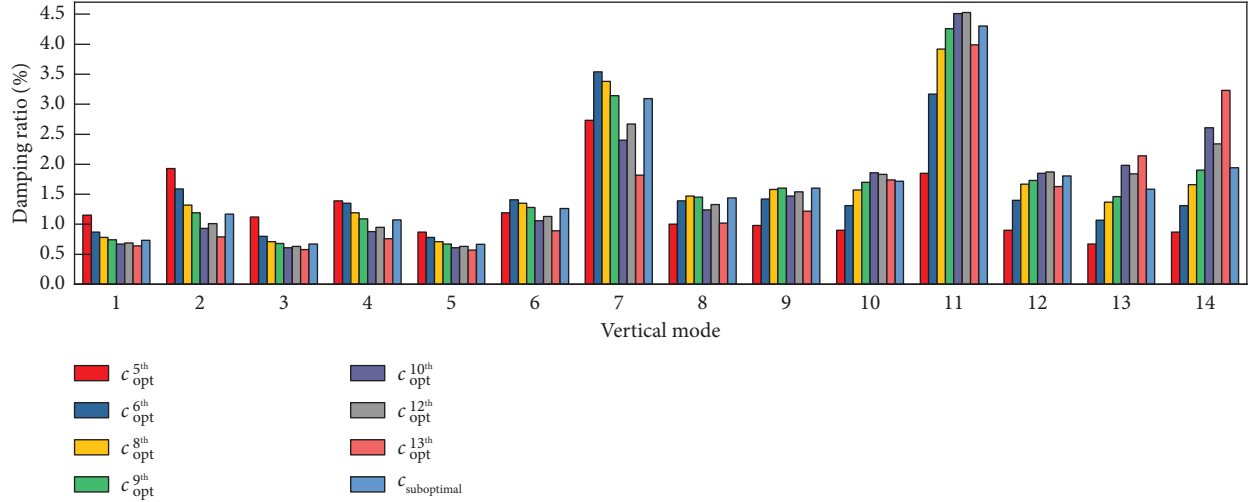


FIGURE 12: Total damping ratios of different damping coefficients.

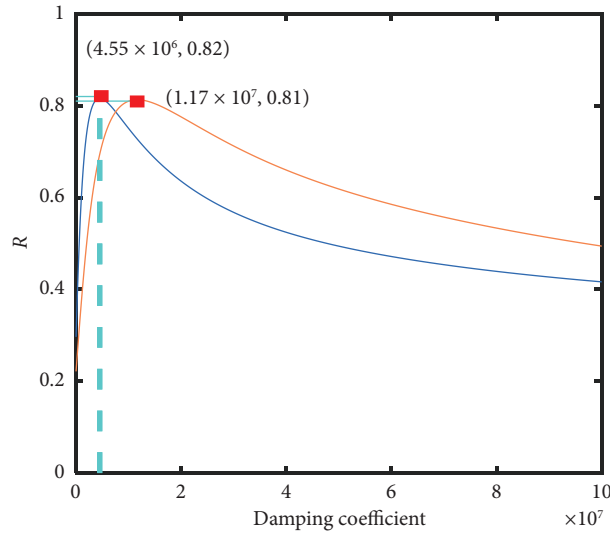


FIGURE 13: Relationship between the damping coefficient of dampers and the target expectation function.

TABLE 7: Major parameters of large-scale 200 kN MR dampers [33].

Parameter	Value
A	2679 m^{-1}
γ, β	647.46 m^{-1}
k_0	$137,810 \text{ N/m}$
N	10
x_0	0.18 m
k_1	617.31 N/m

dampers with a common damping coefficient cannot achieve the satisfactory VIV control performance for each mode of vibration. The SAC-SE control performs better than the passive control owing to its control force can be adjusted to approach the optimal control force. For the 5th modes of vibration, the SAC-SE control can mitigate the VIR amplitudes to 0.190 m, which is below the corresponding threshold value of 0.218 m. For all the rest modes of vibration, the SAC-SE control can reduce the

VIR amplitudes to very small values less than their threshold values in a shorter time. All these results demonstrate that the SAC-SE control is more effective and robust than the passive control in mitigating VIV of the prototype bridge for the selected fourteen vertical modes of vibration.

As shown in Figure 15 and Table 8, in terms of the SAC-SE control, the control effect of the damper-bridge systems 2&3 is better than that of the damper-bridge systems 1&2 in the 1st, 4th, 6th, 8th, and 10th modes of vibration but less than that of the damper-bridge systems 2&3 in the 2nd, 3rd, 5th, and 7th modes of vibration. It is also noted that for the 5th mode of vibration, none of the four control strategies of the damper-bridge systems 2&3 can mitigate the VIR amplitude of the bridge deck to its threshold value. To this end, the following discussion refers to the damper-bridge systems 1&2.

In the bang-bang SAC algorithm, the control force is decided by

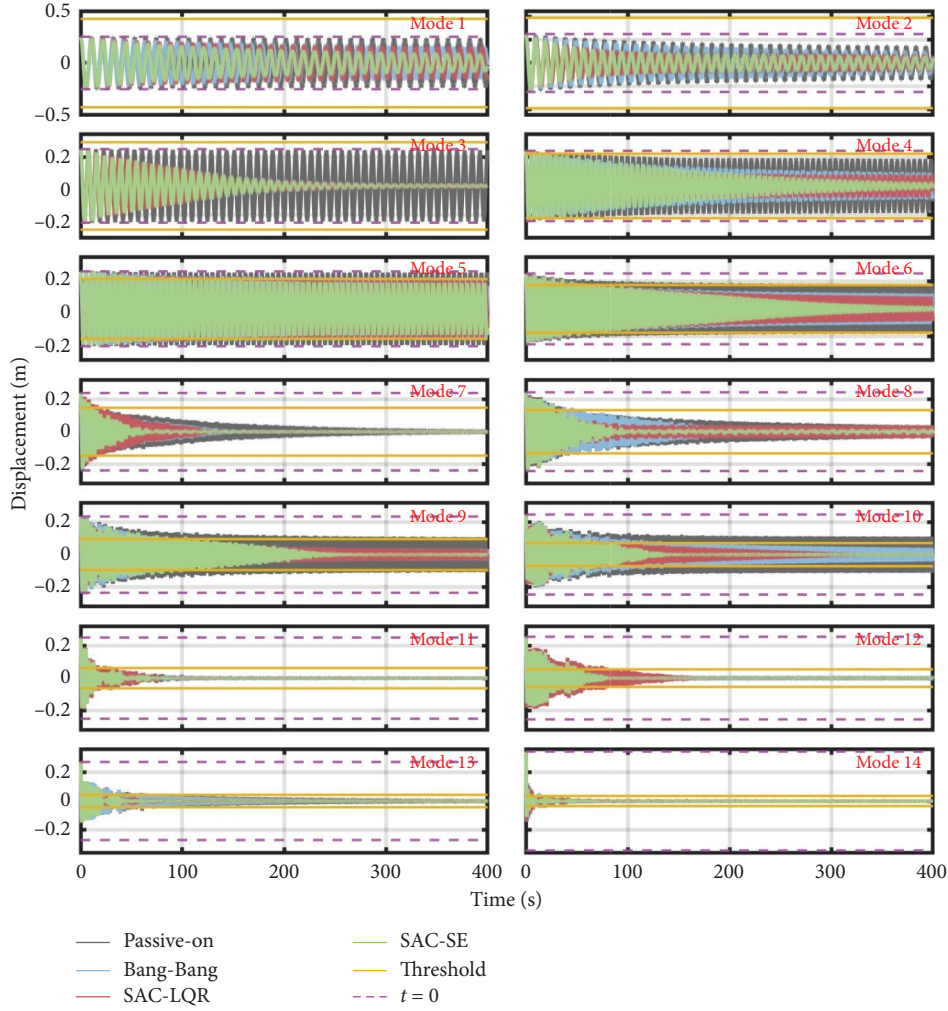


FIGURE 14: MVIV control performance of damper-bridge Systems 1 and 2 with different control strategies.

$$F_{MRD} = \begin{cases} F_{\max}, & F_{MRD}v < 0, \\ F_{\min}, & F_{MRD}v > 0. \end{cases} \quad (18)$$

Compared with the optimal control force decided by equation (14) based on the LQR control algorithm, equation (18) is simple and does not need to calculate the optimal control force using the LQR control algorithm. On the other hand, the SAC-LQR control, like the SAC-SE control, calculates the optimal control force using the LQR algorithm and equation (14). Compared to the proposed SAC strategy in this study, when determining the state space of the bridge, the traditional SAC-LQR control strategy does not involve the linear VIF vector into the state space of the bridge. That is, all the VIFs (linear and nonlinear) are regarded external excitations. Consequently, the state feedback gain matrix of SAC-LQR is determined without considering external excitations. Then, utilizing the state feedback gain matrix and measured velocity and displacement of MR damper, the SAC-LQR control strategy can be implemented. In order to obtain the suitable values of α and β in SAC-SE and SAC-LQR, the value of α remains 100 and the value of β is reduced step by step.

When the control effect is no longer further improved, the values of α and β at this moment are taken as the final values. Consequently, the optimal values of $\alpha_{SAC-LQR}$ and $\beta_{SAC-LQR}$ in SAC-LQR are 100 and 3.2×10^{-4} , respectively. The optimal values of α_{SAC-SE} and β_{SAC-SE} in SAC-SE are 100 and 2.0×10^{-4} , respectively.

As shown in Figure 14, it can be found that the VIR amplitudes are reduced more with the SAC-LQR control than with the bang-bang control and that the time taken for the VIR amplitudes reduced to the corresponding threshold value is shorter for all the fourteen vertical modes of vibration. This is mainly because the control force according to the SAC-LQR control algorithm is the optimal control force, while the bang-bang control is simply based on the relative direction between the control force and damper velocity. Compared to the SAC-LQR control, the SAC-SE control has a better control performance for all the fourteen vertical modes of vibration. This is mainly because the SAC-SE control involves the linear VIF vector into the state space of the bridge to make the decided optimal control force more suitable for control MVIV of a long-span suspension bridge.

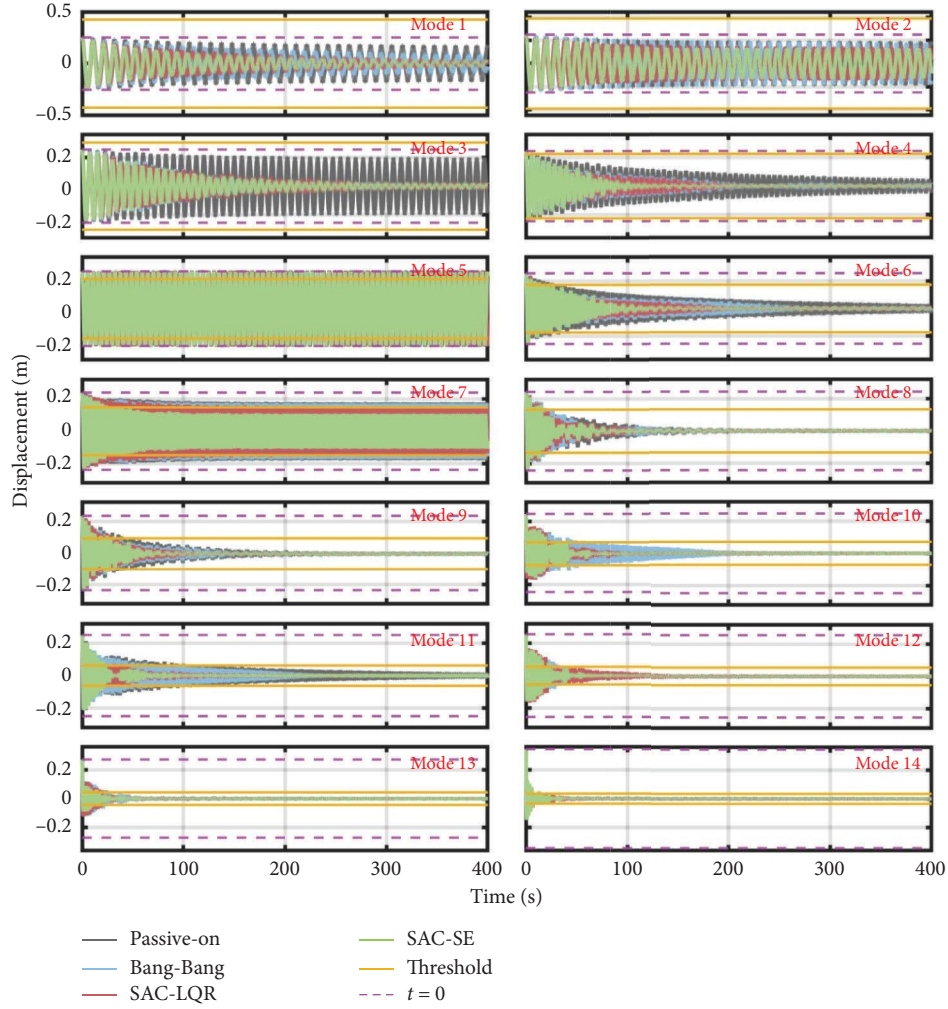


FIGURE 15: MVIV control performance of damper-bridge Systems 2 and 3 with different control strategies.

TABLE 8: The variation of vibration amplitude of the bridge with different control methods.

Vertical mode	$t = 0$ (m)	Damper-bridge Systems 1 and 2				Damper-bridge Systems 2 and 3			
		Passive-on (m)	Bang-Bang (m)	SAC-LQR (m)	SAC-SE (m)	Passive-on (m)	Bang-Bang (m)	SAC-LQR (m)	SAC-SE (m)
1st	0.252	0.236	0.155	0.153	0.088	0.173	0.088	0.029	0.026
2nd	0.25	0.145	0.100	0.065	0.058	0.188	0.178	0.142	0.131
3rd	0.254	0.246	0.009	0.007	0.003	0.197	0.020	0.017	0.006
4th	0.243	0.186	0.090	0.078	0.026	0.049	0.017	0.007	0.006
5th	0.273	0.265	0.228	0.206	0.190	0.272	0.271	0.271	0.270
6th	0.258	0.183	0.111	0.091	0.019	0.034	0.012	0.008	0.006
7th	0.238	0.012	0.001	0.001	0.001	0.178	0.172	0.149	0.107
8th	0.242	0.038	0.013	0.039	0.006	0.006	0.003	0.001	0.001
9th	0.234	0.106	0.033	0.028	0.002	0.007	0.004	0.002	0.002
10th	0.246	0.111	0.041	0.003	0.001	0.002	0.001	0	0
11th	0.249	0.001	0.001	0.001	0	0.017	0.001	0	0
12th	0.254	0.001	0.012	0	0	0.001	0.001	0	0
13th	0.27	0.011	0	0	0	0.001	0	0	0
14th	0.342	0	0	0	0	0	0	0	0

To evaluate the power consumption of different SAC strategies, Figure 16 shows a comparison of the hysteretic curves for 6th mode of vibration among different SAC strategies of the damper-bridge systems 1&2. With the

control system location and the maximum force of the MR damper held constant, it is observed that the SAC-SE strategy exhibits lower power consumption, compared to the other two SAC methods. This demonstrates the higher

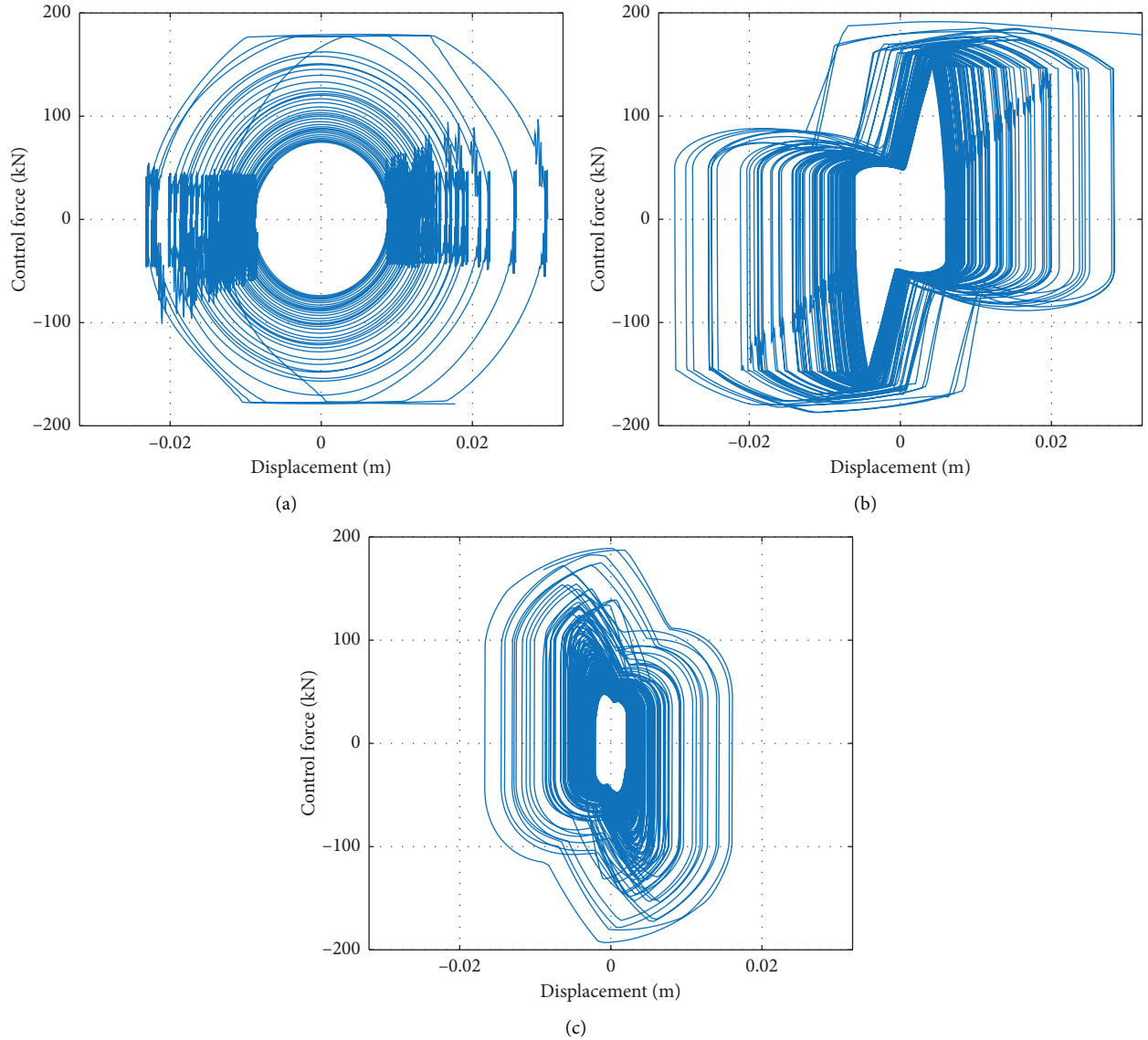


FIGURE 16: Comparison of the hysteretic curves of the 6th mode of vibration among different semiactive control strategies: (a) Bang-Bang; (b) SAC-LQR; (c) SAC-SE.

energy efficiency of the SAC-SE strategy. Consequently, under the same prerequisites for the position of the control system and the maximum force of the MR damper, the SAC-SE control has the better control performance for all the selected fourteen vertical modes of vibration than the other two SAC strategies.

4. Conclusions

A multimodal control strategy for mitigating MVIV of long-span suspension bridge using MR dampers has been developed and applied to a real long-span suspension bridge as a case study. The key findings and conclusions drawn from this study can be summarized as follows:

1. The case study shows that without control, the maximum displacement amplitudes of the prototype bridge deck exceed the threshold values for the 4th to 14th modes of vibration. Therefore, it is necessary to adopt an effective countermeasure to mitigate the MVIV of the prototype bridge.
2. The case study demonstrates that the proposed control strategy performs well and can mitigate the vortex-induced maximum displacement response of the bridge deck to a level less than the threshold value for each of the first fourteen vertical modes of vibration of the bridge deck.
3. In comparison with the passive control, the proposed SAC-SE control demonstrates superior and robust control performance. The passive control using the viscous dampers with a common damping coefficient cannot achieve the satisfactory MVIV control performance for all the first fourteen vertical modes of vibration.

4. The dynamic complex analysis method was adopted to find the additional damping ratios provided by the viscous dampers in different modes of vibration. It was found that the optimal damping coefficients of the dampers leading to the maximum additional damping ratios are different for different modes of vibration, which make the passive control using viscous dampers is difficult to be implemented in practice.
5. A target expectation function was thus proposed to calculate the suboptimal damping coefficient of the dampers for all the fourteen vertical modes of vibration. The suboptimal damping coefficients of the damper-bridge systems 1&2 and the damper-bridge systems 2&3 are 4.55×10^6 N·s/m and 1.17×10^7 N·s/m, respectively.
6. The SAC-SE control has a better control performance than the other two traditional SACs for mitigating MVIV in all the fourteen vertical modes of vibration. The two traditional SACs refer to the bang-bang SAC and the semiactive LQR control.

While this numerical study demonstrates the effectiveness of the proposed SAC algorithm for multimodal control of VIV of a long-span suspension bridge, it will be more exciting to implement it to a real long-span suspension bridge in near future. In the implementation, it should ensure that there are enough anemometers and accelerometers installed on the bridge to measure incoming mean wind speeds and acceleration responses of the bridge deck so that the state feedback matrix can be worked out appropriately according to the field measurement data. Based on the state feedback matrix and the real-time measured velocity and displacement of MR dampers, the optimal control force can be figured out to mitigate VIV of a long-span suspension bridge for a particular mode of vibration. Since all this will be done in real time, it should pay attention to the common issues in real time control applications such as the influence of sampling noise and the time delay of SAC of MR dampers. Furthermore, although this study demonstrates the effectiveness of the proposed SAC algorithm on a specific bridge, the validation and application of the proposed methodology to other types of bridges are desirable.

Data Availability Statement

All data, models, or codes that support the findings of this study are available upon reasonable request.

Disclosure

Any opinions and conclusions presented in this paper are entirely those of the authors.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

S. J. Jiang: investigation, methodology, software, formal analysis, visualization, data curation, and writing—original draft. Y. L. Xu: conceptualization, investigation, methodology, funding acquisition, supervision, and writing—review and editing. G. Q. Zhang: software and data curation. H. Y. Li: software and data curation. S. M. Li: investigation and data curation.

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